

Dirichlet BC's & Constraints

Danube-Tisza Interfluve

$$\text{PDE: } -\nabla \cdot [bK \nabla h] = q_p \quad x \in [0, L]$$

$$\text{BC: } h(0) = h_D \quad h(L) = h_T$$

$\Rightarrow$ heterogeneous (non-zero) Dirichlet BC's

$$K = \text{const.} \ \& \ b = \text{const.} \Rightarrow \text{PDE: } -\nabla^2 h = \frac{q_p}{bK} = f_s$$

Initially lets solve homogeneous problem

$$h(0) = h(L) = 0$$

not realistic but helpful step

Discretize PDE:

$$\underbrace{-\underline{D} \underline{G}}_{\underline{L}} \underline{h} = \underline{f_s}$$

$$\underline{f_s} = \frac{q_p}{bK} \text{ones}(N, 1)$$

$$\Rightarrow \boxed{\underline{L} \underline{h} = \underline{f_s}}$$

$$\underline{L} = -\underline{D} \underline{G} \quad \text{"system matrix"}$$

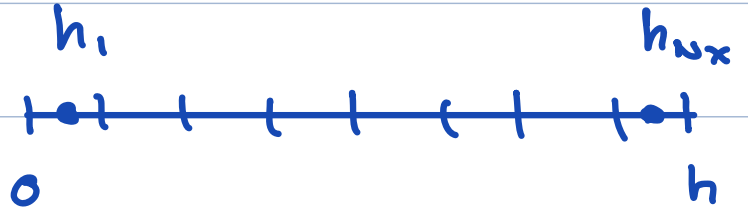
Matlab note: Use backslash to solve lin. sys!

$$\underline{h} = \underline{L} \backslash \underline{f_s} \quad (\text{don't invert } \underline{L})$$

Homogeneous BC's

$$h(0) = h(L) = 0$$

$$\Rightarrow h_1 = 0, h_{N_x} = 0$$



note: the BC are imposed at cell centers

$\Rightarrow$ introduces an error

(in 1D we can just shift grid!)

BC's are Linear $\rightarrow$ write as linear system

$$\underline{B} \times \underline{h} = \underline{0}$$

B = "constraint matrix"

$$N_c \cdot N_x \quad N_x \cdot | \quad N_c \cdot |$$

N_c = number of constraints



Diagram illustrating the relationship between matrices B , h , and G .

Matrix B is an $N \times N_c$ matrix. It has a 1 in the top-left corner and a 1 in the bottom-right corner, with 0s elsewhere.

Matrix h is an $N \times 1$ vector. It contains elements $h_1, h_2, \dots, h_N$.

Matrix G is a $1 \times N_c$ matrix. It has 0s in the first and last positions, and 0s elsewhere.

The equation $B \cdot h = G$ is shown, indicating that the product of matrix B and vector h equals matrix G .

Full discrete problem

$$\text{PDE: } \underline{\underline{L}} \underline{h} = \underline{f_s}$$

$\underline{\underline{L}} = N_x \cdot N_x$ system matrix

$$\text{BC: } \underline{\underline{B}} \underline{h} = \underline{0}$$

$\underline{\underline{B}} = N_c \cdot N_x$ constraint matrix

Neither system has a unique solution
but together they do!

⇒ combine systems by eliminating
constraints in $\underline{\underline{B}}$ from $\underline{\underline{L}}$.

Reduced Linear System

Constraints reduce number of unknowns to $N_x - N_c$

⇒ solve smaller system

reduced system: $\underline{\underline{L_r}} \underline{h_r} = \underline{f_{sr}}$

$\underline{h_r}$ is $N_x - N_c \cdot 1$ reduced solution vector

$\underline{f_{sr}}$ is $N_x - N_c \cdot 1$ reduced r.h.s. vector

$\underline{\underline{L_r}}$ is $(N_x - N_c) \cdot (N_x - N_c)$ reduced system matrix

What is relation between: $\underline{h}_r \longleftrightarrow \underline{h}$
 $\underline{f}_r \longleftrightarrow \underline{f}_s$
 $\underline{L}_r \longleftrightarrow \underline{L}$

Projection matrix

Two vectors of different length are related by a rectangular matrix.

$$\begin{array}{ccccc} \underline{h} & = & \underline{N} & * & \underline{h}_r \\ N \times 1 & & N \times (N - N_c) & & (N - N_c) \times 1 \end{array}$$

What is $\underline{N}$?

For now just require $\underline{N}$ is orthonormal.

$$\underline{N} = \begin{bmatrix} | & | & & | \\ \underline{n}_1 & \underline{n}_2 & \dots & \underline{n}_{N-N_c} \\ | & | & & | \end{bmatrix}$$

$\underline{n}_i$ is the i -th column

$$\underline{n}_i \cdot \underline{n}_i = 1 \quad (\text{normal})$$

$$\underline{n}_i \cdot \underline{n}_{j \neq i} = 0 \quad (\text{ortho})$$

It follows:

$$a) \quad \underline{\underline{N}}^T \underline{\underline{N}} = \underline{\underline{I}}_r$$

$$(N_x - N_c) \cdot N_x \quad N_x \cdot (N_x - N_c) \quad (N_x - N_c) \cdot (N_x - N_c)$$

$$b) \quad \underline{\underline{N}} \underline{\underline{N}}^T = \underline{\underline{I}}'$$

$$N_x \cdot (N_x - N_c) \quad (N_x - N_c) \cdot N_x \quad N_x \cdot N_x$$

$\underline{\underline{I}}_r$ identity in reduced space

$\underline{\underline{I}}'$ "identity in full space but with N_c zeros on the diagonal

if $\underline{\underline{u}} = \underline{\underline{N}} \underline{\underline{u}}_r$

$$\underline{\underline{N}}^T \underline{\underline{u}} = \underline{\underline{N}}^T \underline{\underline{N}} \underline{\underline{u}}_r = \underline{\underline{I}}_r \underline{\underline{u}}_r = \underline{\underline{u}}_r$$

$$\Rightarrow \underline{\underline{u}}_r = \underline{\underline{N}}^T \underline{\underline{u}}$$

$\underline{\underline{N}}$ allows transfer between full & reduced space

$$\begin{aligned} \underline{\underline{u}} &= \underline{\underline{N}} \underline{\underline{u}}_r \\ \underline{\underline{u}}_r &= \underline{\underline{N}}^T \underline{\underline{u}} \end{aligned}$$

we say that $\underline{\underline{N}}^T$ projects $\underline{\underline{u}}$ into reduced space.

Similarly: $\underline{\underline{f}}_s = \underline{\underline{N}} \underline{\underline{f}}_{sr}$

$$\underline{\underline{f}}_{sr} = \underline{\underline{N}}^T \underline{\underline{f}}_s$$

How is $\underline{\underline{L}}$ projected into reduced space?

$$\underline{\underline{L}} \underline{\underline{h}} = \underline{\underline{f}}_s \quad \longrightarrow \quad \underline{\underline{L}}_r \underline{\underline{u}}_r = \underline{\underline{f}}_{sr}$$

already know $\underline{\underline{u}} = \underline{\underline{N}} \underline{\underline{u}}_r$ and $\underline{\underline{f}}_s = \underline{\underline{N}} \underline{\underline{f}}_{sr}$

$$\Rightarrow \underline{\underline{L}} \underline{\underline{N}} \underline{\underline{h}}_r = \underline{\underline{N}} \underline{\underline{f}}_{sr}$$

need to remove $\underline{\underline{N}}$ on l.h.s.

$$\underline{\underline{N}}^T \underline{\underline{L}} \underline{\underline{N}} \underline{\underline{h}}_r = \underbrace{\underline{\underline{N}}^T \underline{\underline{N}}}_{\underline{\underline{I}}_r} \underline{\underline{f}}_{sr} = \underline{\underline{f}}_{sr}$$

$$\underbrace{\underline{\underline{N}}^T \underline{\underline{L}} \underline{\underline{N}}}_{\underline{\underline{L}}_r} \underline{\underline{h}}_r = \underline{\underline{f}}_{sr}$$

Hence we have:

$$\underline{\underline{L}}_r \underline{\underline{h}}_r = \underline{\underline{f}}_{sr} \quad \text{where}$$

$$\underline{\underline{L}}_r = \underline{\underline{N}}^T \underline{\underline{L}} \underline{\underline{N}}$$

$$\underline{\underline{h}}_r = \underline{\underline{N}}^T \underline{\underline{h}}$$

$$\underline{\underline{f}}_{sr} = \underline{\underline{N}}^T \underline{\underline{f}}_s$$

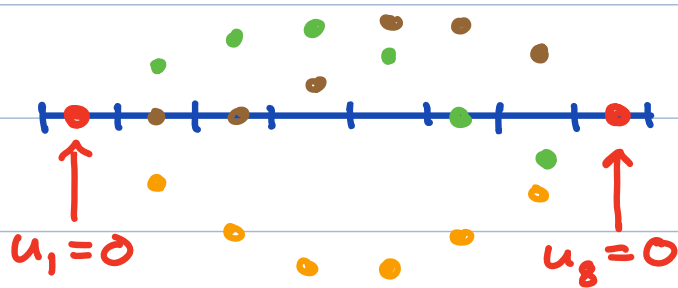
Just need to find $\underline{\underline{N}}$!

$\underline{\underline{N}}$ needs to contain information about BC's.

$\Rightarrow$ related to $\underline{\underline{B}}$

Constraint System: $\underline{\underline{B}} \underline{h} = \underline{0}$

$\Rightarrow$ many solutions



• solu 1: $\underline{h}_1$
• solu 2: $\underline{h}_2$
• solu 3: $\underline{h}_3$ } Null space of $\underline{\underline{B}}: \mathcal{N}(\underline{\underline{B}})$

$\Rightarrow$ search for solutions in $\mathcal{N}(\underline{\underline{B}})$!

$\mathcal{N}(\underline{\underline{B}})$ is the reduced space

Basis of $\mathcal{N}(\underline{\underline{B}})$: Set of linearly independent vectors $\{\underline{n}_i\}$ that span the space.

$$\underline{h}_1 = \sum \alpha_i \underline{n}_i \quad \underline{h}_2 = \sum \beta_i \underline{n}_i \quad \underline{h}_3 = \sum \gamma_i \underline{n}_i$$

$$\Rightarrow \underline{h} = \sum \delta_i \underline{n}_i \quad \text{weights } \alpha_i, \beta_i, \gamma_i, \delta_i \in \mathbb{R}$$

Matrix $\underline{\underline{N}}$ has basis vectors as columns

$$\underline{\underline{N}} = \begin{bmatrix} | & | & | & \dots & | \\ \underline{n}_1 & \underline{n}_2 & \underline{n}_3 & \dots & \underline{n}_{N_x - N_c} \\ | & | & | & \dots & | \end{bmatrix}$$

$$\underline{h} = \sum \delta_i \underline{n}_i = \underline{\underline{N}} \underline{\delta} \quad \underline{\delta} = \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \vdots \\ \delta_{N_x - N_c} \end{bmatrix}$$

$$\underline{h} = \underline{\underline{N}} \underline{h_r} \Rightarrow \underline{h_r} = \underline{\delta} !$$

Note: basis $\{\underline{n}_i\}$ is not unique ∇

simplest orthonormal basis will do

How do we find $\{\underline{n}_i\}$ and hence $\underline{N}$?

In Matlab: $\underline{N} = \text{null}(\underline{B})$

$\underline{N} = \text{spnull}(\underline{B})$

→ download Matlab Central!

Problem: General but expensive.

Simple constraints → $\underline{N}$ easy to find

$$\underline{\underline{B}} = \begin{bmatrix} 1 & & & & & & & \\ & & & & & & & 1 \end{bmatrix}$$

$$\underline{\underline{I}} = \begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

$$\underline{\underline{N}}$$

$$\begin{matrix} & \text{Nx - Nc} \\ & \begin{bmatrix} & & & & & \\ 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix} \\ \text{Nx} & & & & & & & \end{matrix} \quad \underline{\underline{N}}$$

$$\underline{n}_1 \quad \underline{n}_2 \quad \cdot \quad \cdot \quad \cdot \quad \underline{n}_6$$

$\Rightarrow$ splitting $\underline{\underline{I}}$ into $\underline{\underline{B}}$ & $\underline{\underline{N}}$

$\underline{n}_i$'s are the columns of $\underline{\underline{I}}$ not in $\underline{\underline{B}}$
 since each $\underline{n}_i$ has only one non-zero
 entry for one unknown $\Rightarrow$ orthonormal
 $\Rightarrow$ simplest possible basis for $\mathcal{N}(\underline{\underline{B}})$

In Matlab:

1) Vector containing Dirichlet cells:

$$\text{BC.dof_dir} = [\text{Grid.dof_xmin}; \text{Grid.dof_xmax}];$$

2) Build $\underline{\underline{B}}$ selecting rows from $\underline{\underline{I}}$

$$\underline{\underline{B}} = \underline{\underline{I}}(\text{BC.dof_dir}, :);$$

3) Build $\underline{\underline{N}}$ deleting columns from $\underline{\underline{I}}$

$$\underline{\underline{N}} = \underline{\underline{I}};$$

$$\underline{\underline{N}}(:, \text{BC.dof_dir}) = [];$$

Heterogeneous BC's

Return to original geotherm problem

$$\text{PDE: } -\nabla \cdot K \nabla h = q_p \quad x \in [0, L]$$

$$\text{BC: } h(0) = h_b \quad h(L) = h_T$$

Discrete systems

$$\underline{\underline{L}} h = f_s$$

$$\underline{\underline{B}} h = g$$

$$g = \begin{bmatrix} T_s \\ T_b \end{bmatrix}$$

B same as before $\rightarrow$ location of BC's

Because B u = g is linear

$\Rightarrow$ decompose: h = h₀ + h_p

$$\left. \begin{array}{l} \text{homogeneous soln.: } \underline{B} \underline{h}_0 = \underline{0} \\ \text{particular soln.: } \underline{B} \underline{h}_p = \underline{g} \end{array} \right\} \underline{B} (\underbrace{\underline{h}_0 + \underline{h}_p}_{\underline{h}}) = \underline{g}$$

Note: u is unique (assuming suitable BC's)

split into h₀ + h_p is not unique

$\rightarrow$ simplest (obvious) choice

Two questions: 1) How do we find h_p?

2) Given u_p how do we

find associated h₀?

Start with Q2: Suppose we know h_p

$$\underline{L} (\underline{h}_0 + \underline{h}_p) = \underline{f}_s \quad \underline{h}_p \text{ is known} \rightarrow \text{r.h.s}$$

$$\underline{L} \underline{h}_0 = \underbrace{\underline{f}_s - \underline{L} \underline{h}_p}_{\underline{f}_d = \text{Dirichlet r.h.s.}} \quad \underline{L} \underline{h}_0 = \underline{f}_s + \underline{f}_d \quad \underline{f}_d = - \underline{L} \underline{h}_p$$

Form reduced system as before:

$$\underline{L} \underline{h}_{or} = \underline{f}_r \quad \underline{f}_r = \underline{N}^T (\underline{f}_s + \underline{f}_d)$$

where $\underline{h}_{or}$ is reduced homogeneous solution

$$\underline{h}_o = \underline{N} \underline{h}_{or}$$

$$\Rightarrow \underline{h} = \underline{h}_o + \underline{h}_p \quad \text{full solution}$$

Q1: How do we find $\underline{u}_p$?

$\underline{h}_p$ is not unique: $\underline{B} \underline{h}_p = \underline{g}$

$\underline{L} \underline{h}_p \neq \underline{f}_s$ does not satisfy PDE by itself

$\Rightarrow$ just satisfy $\underline{B} \underline{h}_p = \underline{g}$!

Simplest solution: $\underline{h}_p = \begin{bmatrix} h_D \\ 0 \\ 0 \\ \vdots \\ 0 \\ h_T \end{bmatrix}$

$\underline{B} \rightarrow$ location & $\underline{g} \rightarrow$ value

Matlab: $\underline{h}_p(\text{dof_dir}) = \underline{g}$

Simple because our constraints are simple.

General approach for arbitrary constraints

$\Rightarrow$ solve reduced system for $\underline{u}_p$
should be a $N_c \cdot N_c$ system!

$$\begin{array}{ccccc} \underline{h}_{pr} & = & \underline{B} & \underline{h}_p & \rightarrow & \underline{h}_p & = & \underline{B}^T & \underline{h}_{pr} \\ N_c \cdot 1 & & N_c \cdot N_x & N_x \cdot 1 & & N_x \cdot 1 & & N_x \cdot N_c & N_c \cdot 1 \end{array}$$

substitute:

$$\underline{B} \underline{h}_p = \underline{g}$$

$$\underline{B} \underline{B}^T \underline{h}_{pr} = \underline{g}$$

$$N_c \cdot N_x \quad N_x \cdot N_c$$

$$\underline{B} \underline{B}^T$$

$$N_c \cdot N_c$$

Particular reduced system: $\underline{B} \underline{B}^T \underline{h}_r = \underline{g}$

Summary of Dirichlet BC implementation

Two new functions:

1) $[B, N, fn] = \text{build_bnd}(BC, \text{Grid}, I)$

2) $h = \text{solve_lbvp}(L, f, B, g, N)$

$\Rightarrow$ next home work

General implementation

works for 1D, 2D, 3D, cylindrical, spherical, ...

Inside solve_lbrp.m

1) Find $\underline{h}_p$

$$\underline{\underline{B}} \underline{\underline{B}}^T \underline{h}_p = \underline{g} \rightarrow \underline{h}_p = \underline{\underline{B}}^T \underline{h}_{pr}$$

2) Find associated homogeneous solution

$$\underline{\underline{L}}^T \underline{h}_{or} = \underline{f}_r \rightarrow \underline{h}_o = \underline{\underline{N}} \underline{h}_{or}$$

3) Full solution: $\underline{h} = \underline{h}_o + \underline{h}_p$